

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2023 P1H Worked Solutions

Nov 2023 | P1H

29 questions ■ question image followed by worked solution

PREPARED BY

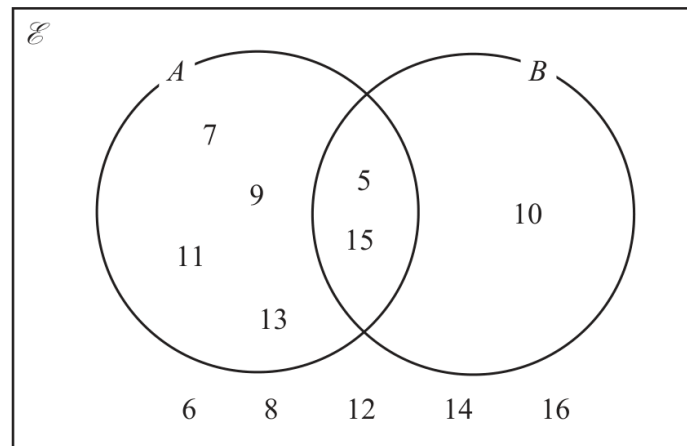
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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

1 Here is a Venn diagram.

List the members of the set

(a) A
(1)(b) $A \cap B$
(1)(c) $(A \cup B)'$
(1)**(Total for Question 1 is 3 marks)**

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Set notation and Venn diagrams. The tag is correct.**Method**

The members of A are all the numbers inside circle A , including the overlap:

$$A = \{5, 7, 9, 11, 13, 15\}$$

The members of $A \cap B$ are the numbers in both circles:

$$A \cap B = \{5, 15\}$$

$(A \cup B)'$ means outside both circles:

$$(A \cup B)' = \{6, 8, 12, 14, 16\}$$

FINAL ANSWER

(a) $\{5, 7, 9, 11, 13, 15\}$, (b) $\{5, 15\}$, (c) $\{6, 8, 12, 14, 16\}$

PAST PAPER QUESTION**Q#: 2****Marks: 6****TOPIC: Forming & Solving Equations**

Nov 2023 P1H - Nov 2023 | P1H

- 2** (a) Factorise fully $12pq - 18p$

.....
(2)

There are 56 metal bars in a box.
Each metal bar is gold or silver or zinc.

y metal bars are gold.

$(3y + 7)$ metal bars are silver.

$(2y - 5)$ metal bars are zinc.

- (b) Work out the number of metal bars that are zinc.
Show clear algebraic working.

.....
(4)

.....
(Total for Question 2 is 6 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 6****TOPIC: Forming & Solving Equations**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to forming and solving equations. The contextual equation carries the main marks.

Solution

(a)

$$12pq - 18p = 6p(2q - 3)$$

(b) The total number of bars is 56, so

$$y + (3y + 7) + (2y - 5) = 56$$

$$6y + 2 = 56$$

$$y = 9$$

The number of zinc bars is

$$2y - 5 = 2(9) - 5 = 13$$

FINAL ANSWER(a) $6p(2q - 3)$, (b) 13 zinc bars

PAST PAPER QUESTION**Q#: 3****Marks: 4**TOPIC: **Percentages**

Nov 2023 P1H - Nov 2023 | P1H

3 Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

.....%

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4**TOPIC: **Percentages**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

Nina pays

$$1500 + 36(450) = 1500 + 16200 = 17700$$

Joshua's profit is

$$17700 - 12500 = 5200$$

Percentage profit:

$$\frac{5200}{12500} \times 100 = 41.6$$

FINAL ANSWER

41.6%

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

Colour	blue	orange	white
Probability	0.58	$2x$	x

- (a) Work out the value of x

$$x = \dots\dots\dots (2)$$

The spinner is spun 250 times.

- (b) Work out an estimate for the number of times the spinner will land on blue.

.....
(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 4**TOPIC: **Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Toolkit. The tag is correct.**Solution**

Let the probability of white be x . Then the probability of orange is $2x$.

$$0.38 + 2x + x = 1$$

$$3x = 0.62$$

$$x \approx 0.207$$

Expected blue sweets:

$$250 \times 0.38 = 95$$

FINAL ANSWER

0.207, and 95.

PAST PAPER QUESTION**Q#: 5****Marks: 3**TOPIC: **Circles, Arcs & Sectors**

Nov 2023 P1H - Nov 2023 | P1H

- 5 The diagram shows a shaded shape made from three identical semicircles, *X*, *Y* and *Z*

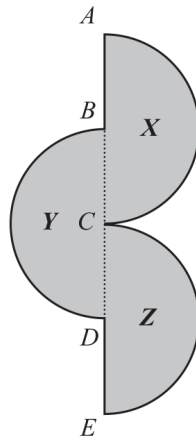


Diagram **NOT**
accurately drawn

ABCDE is a straight line.

AC is the diameter of semicircle *X* and *B* is the centre of semicircle *X*
BD is the diameter of semicircle *Y* and *C* is the centre of semicircle *Y*
CE is the diameter of semicircle *Z* and *D* is the centre of semicircle *Z*

$AC = BD = CE = 20 \text{ cm}$

Work out the perimeter of the shaded shape.
 Give your answer correct to the nearest whole number.

..... cm

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 3****TOPIC: Circles, Arcs & Sectors**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to circles, arcs and sectors.**Solution**

Each semicircle has diameter 20 cm, so each has radius 10 cm.

The total curved perimeter is three semicircles:

$$3 \times \pi r = 3 \times 10\pi = 30\pi$$

The two exposed straight parts are

$$AB = 10, \quad DE = 10$$

So the perimeter is

$$30\pi + 20 = 114.2 \dots$$

FINAL ANSWER

114 cm to the nearest whole number.

PAST PAPER QUESTION**Q#: 6****Marks: 4**TOPIC: **Percentages**

Nov 2023 P1H - Nov 2023 | P1H

- 6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket A $\frac{1}{6}$ off normal price

ticket B 20% off normal price
--

The sale price of ticket **A** is 140 euros.

The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 4**TOPIC: **Percentages**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Percentages. This is a reverse percentage discount comparison.**Method**Ticket A is sold for $\frac{5}{6}$ of its normal price.

$$\text{normal price of A} = 140 \times \frac{6}{5} = 168$$

Ticket B has a 20% discount, so its sale price is 80% of the normal price.

$$\text{normal price of B} = 136 \div 0.80 = 170$$

The difference is

$$170 - 168 = 2$$

FINAL ANSWER

The difference in normal prices is 2 euros.

PAST PAPER QUESTION**Q#: 7****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Nov 2023 P1H - Nov 2023 | P1H

7 $A = 5^3 \times 7^3 \times 11^6$ and $B = 5^6 \times 7^2 \times 11^4$

Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

(Total for Question 7 is 2 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Prime factors, HCF and LCM. The tag is correct.**Method**

$$A = 5^3 \times 7^3 \times 11^6$$

$$B = 5^6 \times 7^2 \times 11^4$$

The HCF takes the smaller power of each common prime:

$$\text{HCF} = 5^3 \times 7^2 \times 11^4$$

FINAL ANSWER

$$5^3 \times 7^2 \times 11^4$$

PAST PAPER QUESTIONTOPIC: **Graphing Inequalities**

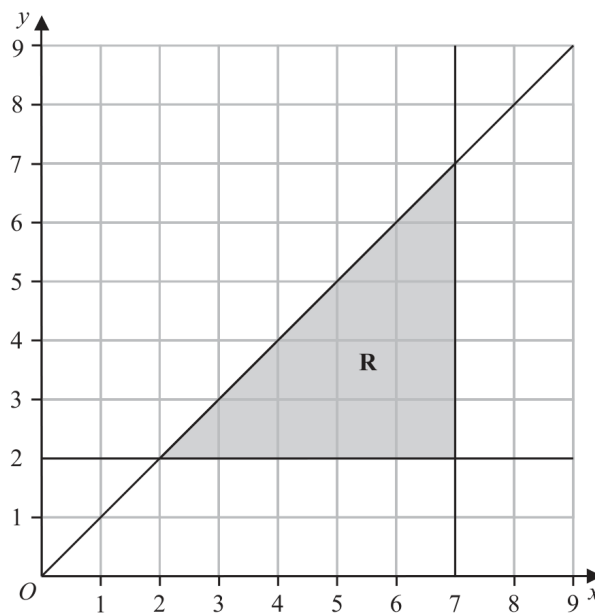
Nov 2023 P1H - Nov 2023 | P1H

Q#: 8**Marks: 5**

- 8 (a) Solve the inequality $8x - 4 \geq 3x - 10$

.....
(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down the three inequalities that define the region **R**

.....
.....
.....
(3)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 5****TOPIC: Graphing Inequalities**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This combines solving and identifying inequalities.**Solution**

For part (a),

$$8x - 4 \geq 3x - 10$$

$$5x \geq -6$$

$$x \geq -\frac{6}{5}$$

For part (b), the region is above $y = 2$, left of $x = 7$, and below $y = x$:

$$y \geq 2$$

$$x \leq 7$$

$$y \leq x$$

FINAL ANSWER(a) $x \geq -\frac{6}{5}$. (b) $y \geq 2$, $x \leq 7$, $y \leq x$.

PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Nov 2023 P1H - Nov 2023 | P1H

- 9 (a) Write 5.87×10^{-4} as an ordinary number.

.....
(1)

- (b) Write 84 000 000 in standard form.

.....
(1)

The number of neurons in a human brain is 8.5×10^{10}

The number of neurons in a monkey brain is 1.47×10^9

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

- (c) Work out the value of K
Give your answer correct to one decimal place.

$K =$
(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Powers, roots and standard form. The tag is correct.**Method**

$$5.87 \times 10^{-4} = 0.000587$$

$$84000000 = 8.4 \times 10^7$$

For part (c),

$$K = \frac{8.5 \times 10^{10}}{1.47 \times 10^9}$$

$$K = \frac{8.5}{1.47} \times 10$$

$$K = 57.823 \dots$$

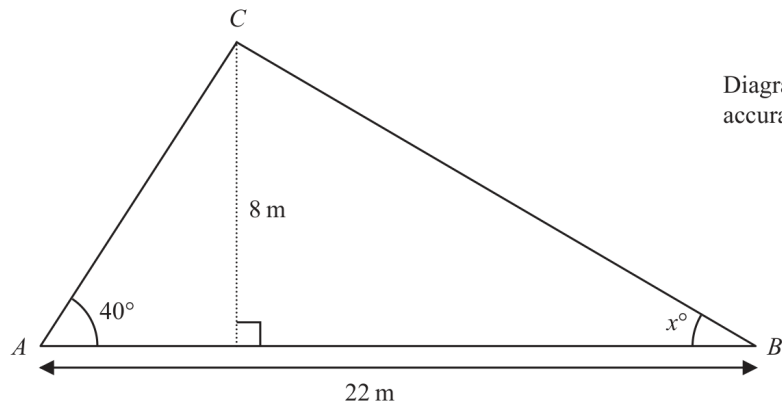
Correct to one decimal place,

$$K = 57.8$$

FINAL ANSWER(a) 0.000587, (b) 8.4×10^7 , (c) 57.8

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2023 P1H - Nov 2023 | P1H

10 Here is triangle ABC Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.
Show your working clearly.

 $x = \dots\dots\dots$ **(Total for Question 10 is 5 marks)**

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Right-Angled Triangles - Pythagoras & Trigonometry.**Solution**Let the perpendicular from C meet AB at F .

$$AF = \frac{8}{\tan 40^\circ} = 9.534 \dots$$

$$FB = 22 - 9.534 \dots = 12.465 \dots$$

In right-angled triangle CFB ,

$$\tan x = \frac{8}{12.465 \dots}$$

$$x = 32.7^\circ$$

FINAL ANSWER

32.7°.

PAST PAPER QUESTION**Q#: 11****Marks: 3**TOPIC: **Algebraic Fractions**

Nov 2023 P1H - Nov 2023 | P1H

- 11 Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms.

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 3**TOPIC: **Algebraic Fractions**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Algebraic Fractions.**Solution**

$$\frac{3}{4} + \frac{5-x}{6x}$$

The common denominator is $12x$:

$$\frac{3}{4} = \frac{9x}{12x}$$

$$\frac{5-x}{6x} = \frac{10-2x}{12x}$$

$$\frac{9x+10-2x}{12x} = \frac{7x+10}{12x}$$

FINAL ANSWER

$$\frac{7x+10}{12x}$$

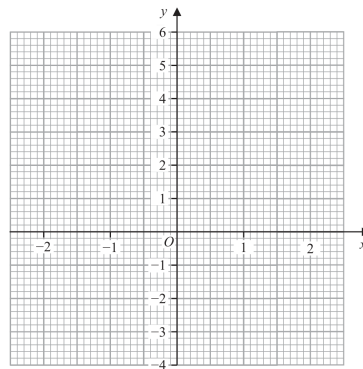
PAST PAPER QUESTIONTOPIC: **Graphs of Functions**

Nov 2023 P1H - Nov 2023 | P1H

Q#: 12**Marks: 5**12 (a) Complete the table of values for $y = x^3 - 3x + 1$

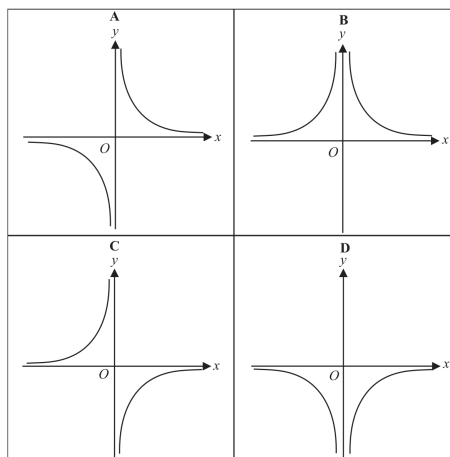
x	-2	-1	0	1	2
y				-1	

(2)

(b) On the grid, draw the graph of $y = x^3 - 3x + 1$ for values of x from -2 to 2

(2)

Here are four graphs.

(c) Write down the letter of the graph that could have the equation $y = -\frac{1}{x^2}$

(1)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Graphs of Functions**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Correct. This is a table-and-graph question for a cubic function.

Solution

$$y = x^3 - 3x + 1$$

The completed table is:

x	-2	-1	0	1	2
y	-1	3	1	-1	3

The graph of

$$y = -\frac{1}{x^2}$$

is below the x -axis on both sides and is symmetrical in the y -axis.

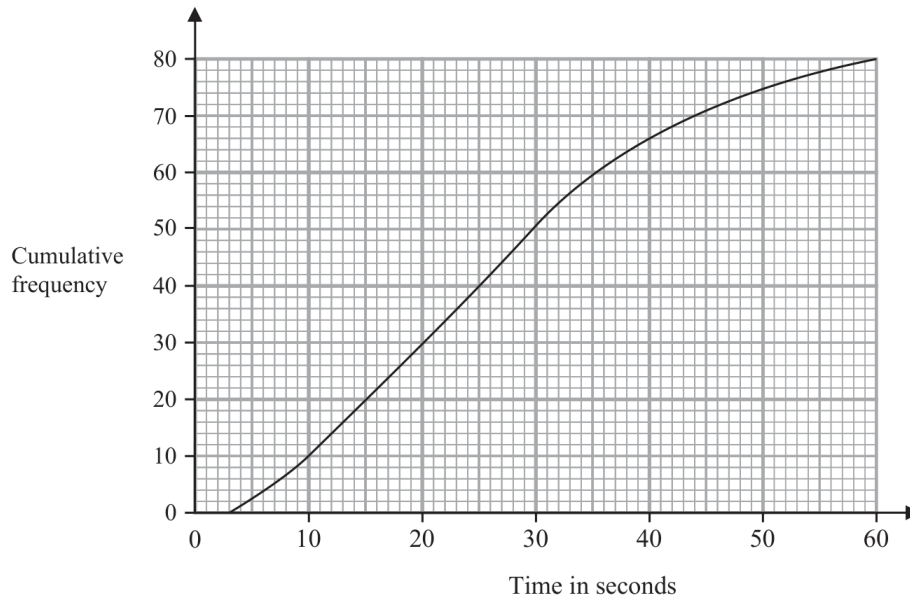
FINAL ANSWER

The graph letter is D.

PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Cumulative Frequency Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

- 13** The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (a) Find an estimate for the median time.

..... seconds
(1)

- (b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

..... %
(3)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 4****TOPIC: Cumulative Frequency Diagrams**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

There are 80 adults, so the median is the 40th value.

From the graph,

$$\text{median} \approx 25 \text{ seconds}$$

At 50 seconds, the cumulative frequency is about 75.

$$80 - 75 = 5$$

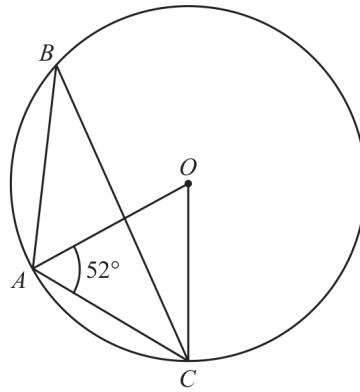
$$\frac{5}{80} \times 100 = 6.25\%$$

FINAL ANSWER

median about 25 seconds, and about 6%.

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Circle Theorems**

Nov 2023 P1H - Nov 2023 | P1H

14Diagram **NOT**
accurately drawn A , B and C are points on a circle, centre O Angle $OAC = 52^\circ$ Find the size of angle ABC

Give reasons for your working.

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 3****TOPIC: Circle Theorems**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**Since $OA = OC$, triangle AOC is isosceles.

$$\angle ACO = \angle OAC = 52^\circ$$

So

$$\angle AOC = 180^\circ - 52^\circ - 52^\circ = 76^\circ$$

The angle at the circumference is half the angle at the centre standing on the same arc:

$$\angle ABC = \frac{76^\circ}{2} = 38^\circ$$

FINAL ANSWER **38° .**

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Rearranging Formulas**

Nov 2023 P1H - Nov 2023 | P1H

15 Make n the subject of the formula $x = \frac{3p+n}{3n-4}$

.....
(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Rearranging Formulas**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rearranging formulas. The tag is correct.**Solution**

$$x = \frac{3p + n}{3n - 4}$$

$$x(3n - 4) = 3p + n$$

$$3xn - 4x = 3p + n$$

$$n(3x - 1) = 3p + 4x$$

FINAL ANSWER

$$n = \frac{3p+4x}{3x-1}$$

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Differentiation**

Nov 2023 P1H - Nov 2023 | P1H

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

.....
(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Differentiation**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to differentiation. The question asks where a curve has a given gradient.

Solution

$$y = 4x^3 - 8x + 5$$

$$\frac{dy}{dx} = 12x^2 - 8$$

The gradient is $\frac{1}{3}$, so

$$12x^2 - 8 = \frac{1}{3}$$

$$12x^2 = \frac{25}{3}$$

$$x^2 = \frac{25}{36}$$

$$x = \frac{5}{6} \quad \text{or} \quad x = -\frac{5}{6}$$

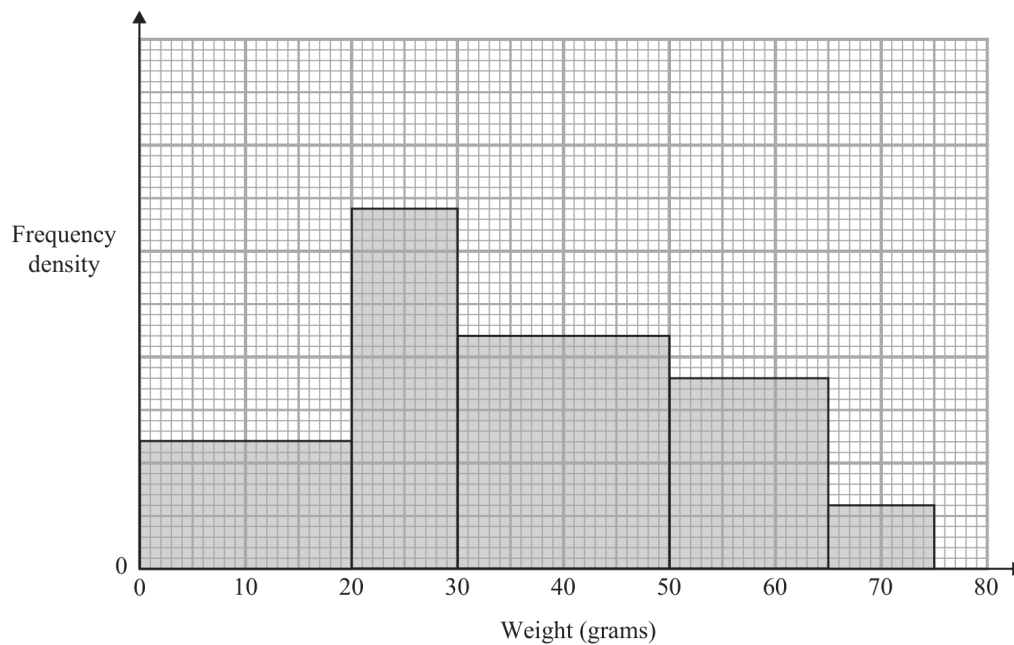
FINAL ANSWER

$$x = \frac{5}{6} \text{ or } x = -\frac{5}{6}.$$

PAST PAPER QUESTION**Q#: 17****Marks: 3**TOPIC: **Histograms**

Nov 2023 P1H - Nov 2023 | P1H

17 The histogram gives information about the weights, in grams, of some oranges in a box.



24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 3**TOPIC: **Histograms**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Histograms. The tag is correct.**Solution**

The 0 to 20 gram class has frequency 24, so its density is

$$\frac{24}{20} = 1.2$$

From the histogram scale, the density for $30 < w \leq 50$ is about 2.2, and the density for $50 < w \leq 65$ is about 1.8.

Medium oranges weigh between 35 g and 55 g:

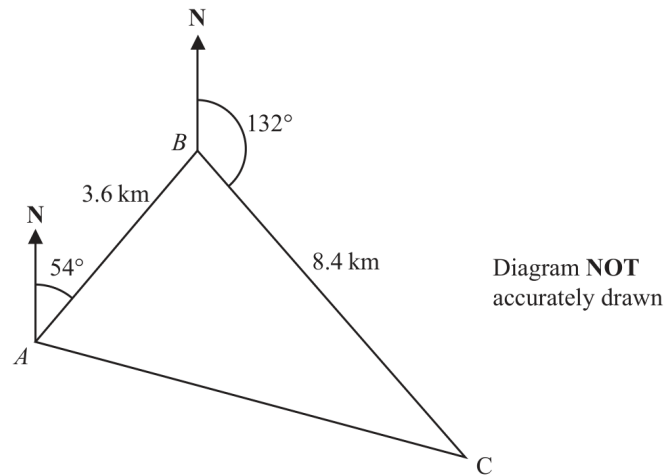
$$15(2.2) + 5(1.8) = 33 + 9 = 42$$

FINAL ANSWER

about 42 medium oranges.

PAST PAPER QUESTION**Q#: 18****Marks: 5****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2023 P1H - Nov 2023 | P1H

18 The diagram shows the positions of three villages, A , B and C The bearing of B from A is 054° The bearing of C from B is 132° Melur walks from A to B She then walks from B to C and from C to A Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.

Give your answer in hours and minutes.

..... hours minutes

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Bearings, Scale Drawing & Constructions. The tag is correct.**Solution**The bearing of B from A is 054° , so the bearing of A from B is

$$054^\circ + 180^\circ = 234^\circ$$

The bearing of C from B is 132° .

So

$$\angle ABC = 234^\circ - 132^\circ = 102^\circ$$

Use the cosine rule to find AC :

$$AC^2 = 3.6^2 + 8.4^2 - 2(3.6)(8.4) \cos 102^\circ$$

$$AC = 9.8028 \dots \text{ km}$$

Total distance walked:

$$3.6 + 8.4 + 9.8028 \dots = 21.8028 \dots \text{ km}$$

At 6 km/h, time is

$$21.8028 \dots \div 6 = 3.6338 \dots \text{ hours}$$

$$0.6338 \dots \times 60 = 38.0 \dots \text{ minutes}$$

FINAL ANSWER

3 hours 38 minutes.

PAST PAPER QUESTION**Q#: 19****Marks: 6**TOPIC: **Sequences**

Nov 2023 P1H - Nov 2023 | P1H

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x =$

(Total for Question 19 is 6 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 6****TOPIC: Sequences**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Sequences. The tag is correct.**Solution**

The sequence is

$$3, 7, 11, 15$$

The common difference is 4, so the n th term is

$$4n - 1$$

The last term is x , so

$$x = 4n - 1$$

$$n = \frac{x + 1}{4}$$

The sum is 7260:

$$\frac{n}{2}(3 + x) = 7260$$

Substitute $n = \frac{x + 1}{4}$:

$$\frac{x + 1}{8}(x + 3) = 7260$$

$$(x + 1)(x + 3) = 58080$$

$$x^2 + 4x - 58077 = 0$$

Factorise:

$$(x - 239)(x + 243) = 0$$

The last term is positive, so

$$x = 239$$

FINAL ANSWER

$$x = 239.$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved to Probability Toolkit.**Method**Let the probability of taking a 10 cent coin be p .

The mean is 10 cents only when both coins are 10 cent coins.

$$p^2 = \frac{49}{121}$$

$$p = \frac{7}{11}$$

So the probability of taking a 20 cent coin is

$$1 - \frac{7}{11} = \frac{4}{11}$$

The mean is 15 cents when one coin is 10 cents and the other is 20 cents.

There are two possible orders:

$$P(10, 20) + P(20, 10)$$

$$= \frac{7}{11} \cdot \frac{4}{11} + \frac{4}{11} \cdot \frac{7}{11}$$

$$= \frac{56}{121}$$

FINAL ANSWER

$$\frac{56}{121}$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Probability Toolkit**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved to Probability Toolkit.**Method**Let the probability of taking a 10 cent coin be p .

The mean is 10 cents only when both coins are 10 cent coins.

$$p^2 = \frac{49}{121}$$

$$p = \frac{7}{11}$$

So the probability of taking a 20 cent coin is

$$1 - \frac{7}{11} = \frac{4}{11}$$

The mean is 15 cents when one coin is 10 cents and the other is 20 cents.

There are two possible orders:

$$P(10, 20) + P(20, 10)$$

$$= \frac{7}{11} \cdot \frac{4}{11} + \frac{4}{11} \cdot \frac{7}{11}$$

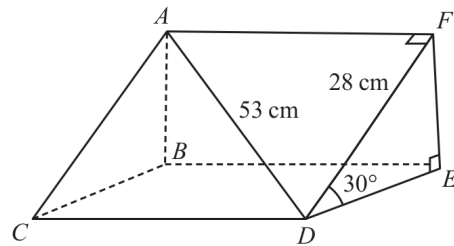
$$= \frac{56}{121}$$

FINAL ANSWER

$$\frac{56}{121}$$

PAST PAPER QUESTION**Q#: 21****Marks: 5**TOPIC: **Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

$AD = 53 \text{ cm}$

$DF = 28 \text{ cm}$

$\text{Angle } FDE = 30^\circ$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3 **(Total for Question 21 is 5 marks)**

PAST PAPER SOLUTION**Q#: 21****Marks: 5**TOPIC: **Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The triangular cross-section is triangle DEF .It is right-angled at E , with

$$DF = 28$$

and

$$\angle FDE = 30^\circ$$

So

$$FE = 28 \sin 30^\circ = 14$$

and

$$DE = 28 \cos 30^\circ = 14\sqrt{3}$$

Area of triangle DEF :

$$\frac{1}{2}(14)(14\sqrt{3}) = 98\sqrt{3}$$

The length of the prism is

$$AD = 53$$

Volume of the prism:

$$98\sqrt{3} \times 53$$

$$= 5194\sqrt{3}$$

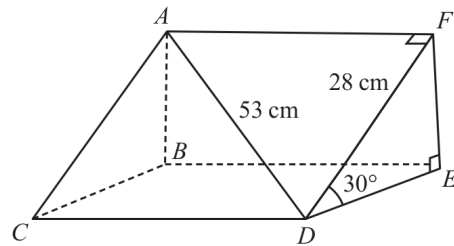
$$= 8996.2719 \dots$$

FINAL ANSWER

$$8996 \text{ cm}^3$$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

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$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3 **(Total for Question 21 is 5 marks)**

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Volume & Surface Area**

Nov 2023 P1H - Nov 2023 | P1H

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The length of the prism is

$$AD = 53$$

Volume of the prism:

$$98\sqrt{3} \times 53$$

$$= 5194\sqrt{3}$$

$$= 8996.2719 \dots$$

FINAL ANSWER

$$8996 \text{ cm}^3$$

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2023 P1H - Nov 2023 | P1H

- 22** [In this question 1 cm = 1 unit on the x -axis and 1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$
 The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P
 L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ
 Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The centre of the circle is $(0, 0)$, and

$$P = (8, -10)$$

The gradient of OP is

$$\frac{-10}{8} = -\frac{5}{4}$$

So the gradient of the tangent is

$$\frac{4}{5}$$

The tangent through P is

$$y + 10 = \frac{4}{5}(x - 8)$$

$$y = \frac{4}{5}x - \frac{82}{5}$$

At the x -axis, $y = 0$:

$$0 = \frac{4}{5}x - \frac{82}{5}$$

$$x = \frac{41}{2}$$

So

$$Q = \left(\frac{41}{2}, 0\right)$$

At the y -axis, $x = 0$:

$$R = \left(0, -\frac{82}{5}\right)$$

Therefore

$$RQ = \sqrt{\left(\frac{41}{2}\right)^2 + \left(\frac{82}{5}\right)^2}$$

$$RQ = 26.252\dots$$

FINAL ANSWER

26.3 cm

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2023 P1H - Nov 2023 | P1H

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 L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ
 Give your answer correct to 3 significant figures.

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(Total for Question 22 is 6 marks)

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Nov 2023 P1H - Nov 2023 | P1H

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FINAL ANSWER

26.3 cm

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Nov 2023 P1H - Nov 2023 | P1H

23 Solid A is similar to solid B

Here is some information about solid A and solid B

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid B
Give your answer as a decimal.

_____ cm

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The area ratio of solid A to solid B is

$$7776 : 486 = 16 : 1$$

So the linear ratio A:B is

$$\sqrt{16} : 1 = 4 : 1$$

Therefore the volume ratio A:B is

$$4^3 : 1^3 = 64 : 1$$

The volumes are 8^x and 2^{x+4} , so

$$\frac{8^x}{2^{x+4}} = 64$$

Write 8^x as a power of 2:

$$8^x = (2^3)^x = 2^{3x}$$

So

$$\frac{2^{3x}}{2^{x+4}} = 2^6$$

$$2^{3x-(x+4)} = 2^6$$

$$2x - 4 = 6$$

$$x = 5$$

The height of solid A is

$$3^x = 3^5 = 243$$

Since the linear ratio A:B is 4 : 1, the height of solid B is

$$\frac{243}{4} = 60.75$$

FINAL ANSWER

60.75 cm

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Nov 2023 P1H - Nov 2023 | P1H

23 Solid A is similar to solid B

Here is some information about solid A and solid B

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid B
Give your answer as a decimal.

_____ cm

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Nov 2023 P1H - Nov 2023 | P1H

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Since the linear ratio A:B is 4 : 1, the height of solid B is

$$\frac{243}{4} = 60.75$$

FINAL ANSWER

60.75 cm

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Transformations of Graphs**

Nov 2023 P1H - Nov 2023 | P1H

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

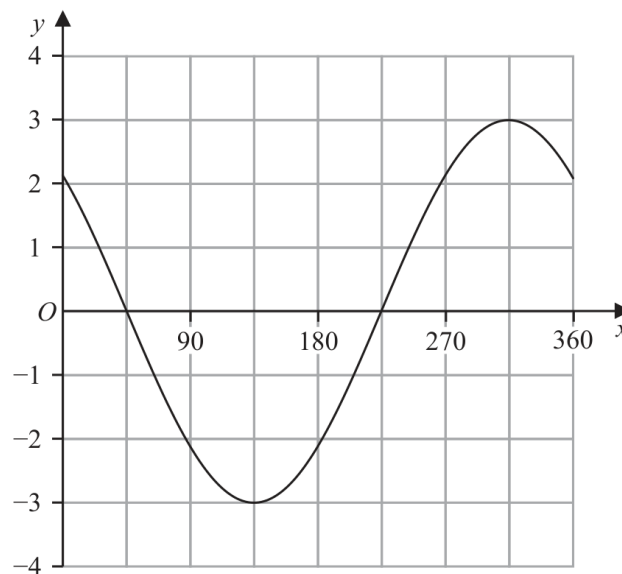
(a) Write down the value of a and the value of b

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(2)

The graph of $y = p \cos(x+q)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$$p = \dots\dots\dots$$

$$q = \dots\dots\dots$$

(2)

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 4****TOPIC: Transformations of Graphs**

Nov 2023 P1H - Nov 2023 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Functions to Transformations of Graphs.**Method**

For part (a),

$$f(x) = 5x^2 + 9x + 2$$

$$g(x) = 5(x + 4)^2 + 9(x + 4) + 8$$

Since

$$f(x + 4) = 5(x + 4)^2 + 9(x + 4) + 2$$

$$g(x) = f(x + 4) + 6$$

So the graph has moved 4 units left and 6 units up.

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$$

For part (b), the graph has maximum value 3 and minimum value -3 , so

$$p = 3$$

The minimum occurs when

$$x + q = 180$$

From the graph, the minimum is at $x = 135$, so

$$135 + q = 180$$

$$q = 45$$

FINAL ANSWER

$$a = -4, b = 6, p = 3, q = 45$$

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Transformations of Graphs**

Nov 2023 P1H - Nov 2023 | P1H

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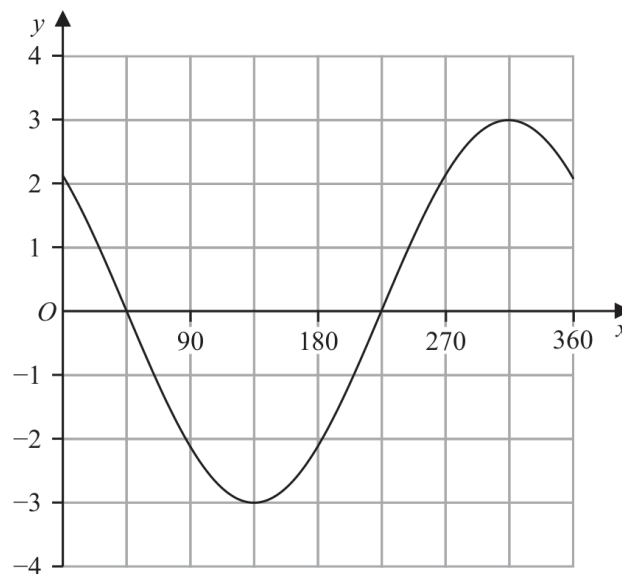
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$a = \dots\dots\dots$

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Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(2)

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 4****TOPIC: Transformations of Graphs**

Nov 2023 P1H - Nov 2023 | P1H

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For part (a),

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Since

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FINAL ANSWER

$$a = -4, b = 6, p = 3, q = 45$$